Bureau
International des
Poids et
Mesures

# Acronyms definitions 1.1th

edition 2019

**Bureau International**
**des Poids et Mesures**

**1.1th edition 2019**

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# Acronyms and definitions

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# 1. Purpose

The purpose of this document is to provide a list of acronyms and definitions employed in the procedures of the Joint Committee for Traceability in Laboratory Medicine (JCTLM).

# 2. Scope

The scope of this document is all procedures that describe the activities of the JCTLM, notably those that are specified as the responsibility of the Database Working Group, the Secretariat or the Executive Committee.

# 3. Acronyms

| | |
|---|---|
| BIPM | International Bureau of Weights and Measures, Website: https://www.bipm.org |
| CIPM | International Committee for Weights and Measures |
| CIPM MRA | The CIPM Mutual Recognition Arrangement |
| CRM | Certified Reference Material |
| DB WG | Database Working Group of the JCTLM, Website: https://www.bipm.org/en/committees/cc/wg/jctlm-dbwg.html |
| DB WG RT | Review Team of the Database Working Group |
| DB WG RTL | Review Team Leader of the Database Working Group |
| ICSH | International Council for standardization in Haematology Website: https://icsh.org/ |
| IFCC | International Federation for Clinical Chemistry and Laboratory Medicine, Website: https://www.ifcc.org |
| ILAC | International Laboratory Accreditation Cooperation, Website: https://www.ilac.org/ |
| ISO | International Organization for Standardization, Website : https://www.iso.org/ |
| IVD | *In Vitro* Diagnostic |
| IVDD | Directive 98/79/EC of the European Parliament and of the Council of 27 October 1998 on *in vitro* diagnostic medical devices |
| JCTLM | Joint Committee for Traceability in Laboratory Medicine, Website : https://www.bipm.org/en/committees/jc/jctlm/ |
| KCDB | The BIPM key comparison database, Website : https://kcdb.bipm.org/ |
| RELA | IFCC External Quality assessment scheme for Reference Laboratories in Laboratory Medicine, Website : https://www.dgkl-rfb.de:81/index.shtml |
| RM | Reference Material |
| RMM | Reference Measurement Method |
| RMP | Reference Measurement Procedure |
| RMM/P | The concatenation of RMM and RMP for brevity in the DB WG procedure documents |
| RML | Reference Measurement Laboratory |
| SI | The International System of Units |
| TEP WG | Working Group on Traceability: Education and Promotion, website: https://www.bipm.org/en/committees/cc/wg/jctlm-wg-tep.html |

VIM International Vocabulary of Metrology

# 4. Definitions

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| Certified Reference Material CRM | reference material, accompanied by documentation issued by an authoritative body and providing one or more specified property values with associated uncertainties and traceabilities, using valid procedures, VIM 3rd Ed., 5.14 (2012). |
| Commutability of a reference material | Property of a reference material, demonstrated by the closeness of agreement between the relation among the measurement results for a stated quantity in this material, obtained according to two given measurement procedures, and the relation obtained among the measurement results for other specified materials, VIM 3rd Ed. , 5.15 (2012). An example is provided in the Attachment 1 of this procedure, Chart JCTLM-0001.0.<br><br>Demonstrating commutability among CRMs with any given measurement process does not assure commutability of any CRM across different measurement processes. |
| Consensus | Group solidarity in sentiment and belief (opinion); operationally, the absence of sustained opposition. |
| Extent of equivalence | An indication of the agreement among measured values of the same quantity assigned to two or more CRMs or ability of different measurement procedures to produce consistent values when used to measure the amount of substance in any given CRM.<br><br>The extent of equivalence can be usefully communicated with Youden or Bland-Altman style graphics that include an indication of measurement uncertainty to identify and place differences among the measured values in perspective. |
| Higher order | The term “higher-order” was left undefined in the IVDD; however, ISO 15193:2009 and ISO 15194:2009 describe the essential requirements for higher-order reference materials and methods. |
| ISO Standards | Normative standards employed by JCTLM in reviewing and judging suitability for listing materials (ISO 15194), methods (ISO 15193) and procedure-defined measurands (ISO 18153) as being of a higher metrological order (ISO 17511) as required in the European Community In Vitro Diagnostic Directive (EC IVDD) (98/79/EC, Annex1 (A) (3) 2$^{nd}$ paragraph) and reference measurement service laboratories (ISO 15195, ISO/IEC 17025:2005). |
| JCTLM Criteria | Reviewing criteria derived from the applicable international standards for certified reference materials, reference measurement procedures and reference measurement services. Primary standards are from the International Organization for Standardization (ISO). |
| JCTLM Database | Database of available higher order reference materials, reference measurement methods/procedures and of reference measurement services provided by reference laboratories that are compliant with the JCTLM criteria, website: https://www.bipm.org/jctlm/ |

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| JCTLM Database WG Chair | Leader of Database WG. The Chair position of the Database WG is held by the Chairman of JCTLM. |
| JCTLM Database WG vice-chair | Responsible for an Analyte Group comprising three or more review teams.<br><br>The composition of each of the three Analyte Groups and their respective Database WG vice-chairs can be identified on the website at https://www.bipm.org/en/committees/cc/wg/jctlm-dbwg.html |
| JCTLM Executive Committee | The Executive Committee is the impartial final decision-making organ, only accountable to the Executive Committee Member Organizations.<br><br>It comprises representatives of the Executive Committee Member Organizations that currently are the JCTLM Founding Organizations and the ICSH. Members of the Executive Committee can be identified on the website: https://www.bipm.org/en/committees/cc/wg/jctlm-exec.html |
| JCTLM Founding Organizations | The three organizations that by a Declaration of Cooperation formed the JCTLM; the BIPM, the IFCC and the ILAC. |
| JCTLM Executive Committee Member Organizations | Intergovernmental and international non-governmental organizations and bodies having technical competence in the field or a subspecialty, that:<br>1. are representative of the specialized field of interest in which they operate;<br>2. are concerned with matters covering a part or all of the Committee's activities;<br>3. have a permanent directing body, authorized representatives and systematic procedures for communicating with its membership. |
| JCTLM National and Regional Members | National and regional organizations that adhere to and/or contribute to the activities of the intergovernmental and international non-governmental organizations that are members of the JCTLM Executive Committee and that have expertise in traceability in laboratory medicine and demonstrate a willingness to provide experts for JCTLM Working Groups and Review Teams. |
| JCTLM Stakeholder Members | Properly constituted "non-profit" and "for-profit" organizations, with interest, expertise and a demonstrable record of working to reduce the between method variability in laboratory medicine measurements and a commitment to promote the JCTLM database and activities. |
| JCTLM Secretariat | Secretariat maintained on behalf of JCTLM by the BIPM, email address: jctlm@bipm.org |
| List I | Certified reference materials and reference measurement methods for well-defined chemical entities or internationally recognized reference method-defined measurands. Reference materials and |

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| | measurement methods included in this category are those that provide values that are traceable to the SI units; e.g., electrolytes, enzymes, drugs, metabolites and substrates, non-peptide hormones, and some proteins. |
| List II | Reference materials for which values of the measurands are not SI-traceable but are assigned by or traceable to an internationally agreed upon protocol, e.g., reference materials for blood typing, coagulation factors, infectious diseases, nucleic acids, and some proteins and purified substances.. List II also contains a group of purified substances which, due to the absence of reference measurement procedures, should not be directly used for calibration of routine methods unless commutability is established and/or matrix effect independent internationally recognized standardized value transfer protocols to commutable samples are applied. |
| List III | Certified Reference Materials for nominal properties |
| Measurand | quantity intended to be measured, VIM 3rd Ed., 2.3 (2012). |
| Measurement principle | phenomenon serving as a basis of a measurement, VIM 3rd Ed., 2.4 (2012). |
| Measurement method | generic description of a logical organization of operations used in a measurement, VIM 3rd Ed., 2.5 (2012). |
| Reference measurement procedure | Measurement procedure accepted as providing measurement results fit for their intended use in assessing measurement trueness of measured quantity values obtained from other measurement procedures for quantities of the same kind, in calibration, or in characterizing reference materials, VIM 3rd Ed., 2.7 (2012) |
| Reference Measurement Laboratory | A laboratory that meets the requirements specified in ISO 15195 as a calibration laboratory. Reference measurement laboratories should implement reference measurement procedures and produce results of measurement that are accurate and traceable to national or international primary reference materials when such are available. Whenever possible, traceability should be established to a reference material which forms an embodiment of the SI unit (ISO 17511).<br><br>This International Standard may form a basis for the accreditation of a reference measurement laboratory that applies for official recognition of the performance of a reference measurement procedure. Reference measurement laboratories are usually accredited by national accrediting bodies. |
| RELA Advisor | Qualified individual appointed by the Executive Committee to assist the Database WG to review the services nominated for assessment by JCTLM and/or listed in the Database. |

# 5. Revision History

| Version number | Date of Issue/Review | Summary of change |
|---|---|---|
| 0.1 | 10/05/2016 | First draft |
| 1.0 | 27/07/2017 | Final version published |
| 1.1 | 18/12/2019 | Editorial modifications |

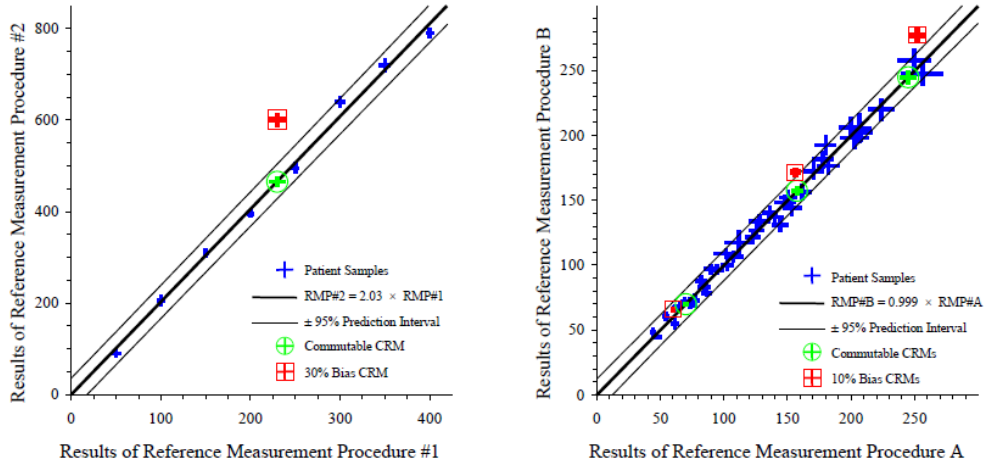
# Annex 1. Operational Definition of Commutability

Chart Example illustrating the distinguishing difference between a commutable and a non-commutable reference material in two measurement procedures:

Step 1 A series of patient samples, selected to cover the analytical range of the methods, are measured using both procedures. The results are plotted on a scatter-graph and the mathematical relationship between the patient sample results from the two procedures established along with a stated confidence interval on that relationship.

Step 2 The substance amount of the measurand in the certified reference material is measured using the same two procedures. Values from commutable CRMs will lie within the confidence interval found for the patient sample with approximately the same stated confidence. Values from non-commutable materials will lie outside the confidence interval.

WG1-(0001.0 Two Graphical Examples of Commutability Evaluations




Graphs are taken from the presentation given at the JCTLM symposium, Paris, June 2002, by Heinz Schimmel, Institute for Reference Materials and Measurements (Left side) and from Richard R. Miller, Dade Behring using data from Table A2, Clinical and Laboratory Standards Institute, EP9-A2. Confidence interval calculations and formatted graphs were provided by David L. Duewer, National Institute of Standards and Technology.

# Related documents


[1] SI *The International System of Units* (SI), 8th Edition, Paris, France (2006). Website: https://www.bipm.org/en/publications/si-brochure/

[2] VIM *International Vocabulary of Metrology—Basic and General Concepts and Associated Terms*, (VIM 3rd edition), JCGM 200:2012 (JCGM 200:2008 with minor corrections) Website: https://www.bipm.org/en/publications/guides/#vim

[3] IVDD Directive 98/79/EC of the European Parliament and of the Council of 27 October 1998 on in vitro diagnostic medical devices, Website.

[4] ISO 15193:2009 In vitro diagnostic medical devices—Measurement of quantities in samples of biological origin—Requirements for content and presentation of reference measurement procedures

[5] ISO 15194:2009 In vitro diagnostic medical devices—Measurement of quantities in samples of biological origin—Requirements for certified reference materials and content of supporting documentation.

[6] ISO 15195:2003 Laboratory medicine—Requirements for reference measurement laboratories.

[7] ISO/IEC 17025:2005 General requirements for the competence of testing and calibration laboratories

[8] ISO 17511:2003 In vitro diagnostic medical devices—Measurement of quantities in biological samples—Metrological traceability of values assigned to calibrators and control materials.

[9] ISO 18153:2003 In vitro diagnostic medical devices—Measurement of quantities in biological samples—Metrological traceability of values for catalytic concentration of enzymes assigned calibrators and control materials.

[10] JCTLM ***Declaration of Cooperation*** between the BIPM, IFCC and ILAC, for the establishment of a Joint Committee for Traceability in Laboratory Medicine (JCTLM), revised in March 2016—available at: https://www.bipm.org/en/worldwide-metrology/jctlm-cooperation/


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